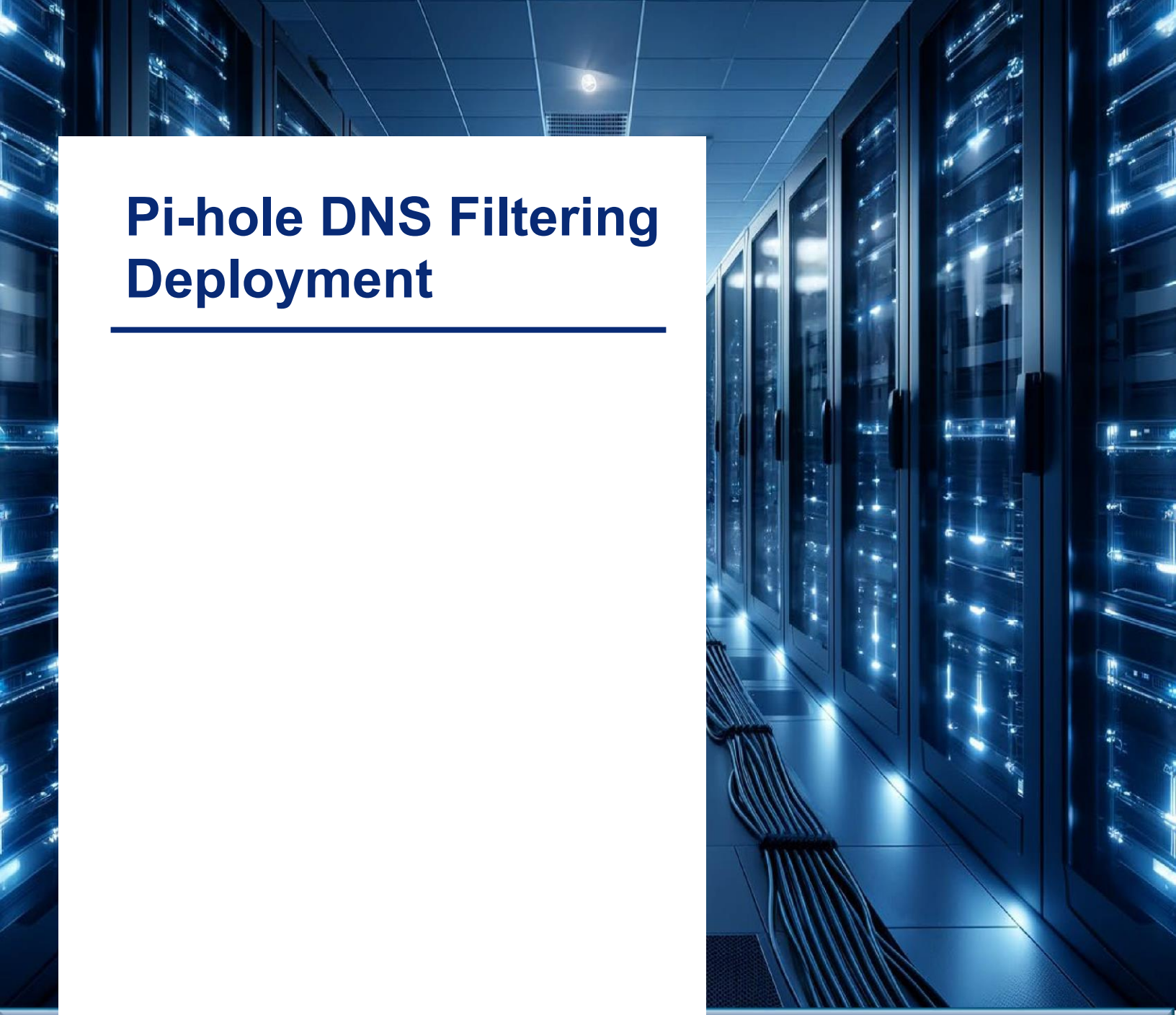

# Pi-hole DNS Filtering Deployment

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# Introduction

## Context

Previously, I was using the Eero Plus service for ad-blocking and content filtering. While convenient, it provided minimal transparency and customization over DNS queries or network filtering. I wanted more control, visibility, and security over DNS resolution across my home network.

## Purpose

The goal of this project was to implement a more advanced and self-managed DNS filtering solution by deploying Pi-hole on a Raspberry Pi. This setup would allow for:

- Network-wide ad blocking
- DNS query logging
- Client-specific filtering policies
- Better understanding of how DNS impacts network security and performance

# Equipment

## Network Equipment and Software

| Hardware Component | Description |
|---|---|
| Raspberry Pi 4 Model B (8GB RAM) | Main compute device running Pi-hole |
| Argon ONE M.2 Case | Aluminum case with passive cooling |
| SanDisk 64GB microSD | Storage medium for OS and Pi-hole |
| Smraza 5.1V 3A Power Supply | Reliable USB-C power source |
| Ethernet Cable | Ensures wired connectivity |

| Software Component | Description |
|---|---|
| Pi-hole v6.x Series | Core: v6.1.4<br>FTL: v6.2.3<br>Web interface: v6.2.1 |
| Raspberry Pi OS Lite (64-bit) | Headless Raspberry Pi deployments |
| Web Admin Interface | Web dashboard managing and monitoring |
| SSH | Remote terminal access |
| Upstream DNS | Cloudflare 1.1.1.1 |

# Configuration and Setup Process

## Hardware Setup

- **Case Assembly:** Installed the Raspberry Pi 4 Model B into the Argon ONE M.2 case, ensuring proper alignment for passive cooling and cable routing.
- **Storage Preparation:** Flashed Raspberry Pi OS Lite (64-bit) to a SanDisk 64GB microSD card using Raspberry Pi Imager.
- **Configuration during Flashing:**
    - Set the hostname
    - Created a user account with a secure password
    - Disabled Wi-Fi to ensure wired-only operation
    - Enabled SSH using public key authentication (private key remains on personal device; public key was injected during flash)
- **Deployment:** Inserted the microSD card, assembled the case, and connected:
    - Power via USB-C (5.1V 3A)
    - Ethernet from the backhaul Eero node directly to the Pi's Ethernet port

## Eero Network Configuration

- **IP Reservation:** Reserved a static IP for the Raspberry Pi in the Eero app under device settings
- **DNS Customization:**
    - Disabled Local DNS Caching in Eero
    - Changed custom DNS to point to the Pi-hole's reserved IP address instead of Cloudflare directly (i.e., all client queries now flow through Pi-hole first)

# Configuration and Setup Process

## Pi-hole Installation

- **Remote Access:** SSH'd into the Pi using the private key (from Mac) for passwordless and secure access
- **Software Installation**: Ran the official Pi-hole installation script:

  `curl -sSL https://install.pi-hole.net | bash`
- **Setup Wizard:**
  - Assigned the static IP (matching the reserved IP in Eero)
  - Received the admin panel password on completion

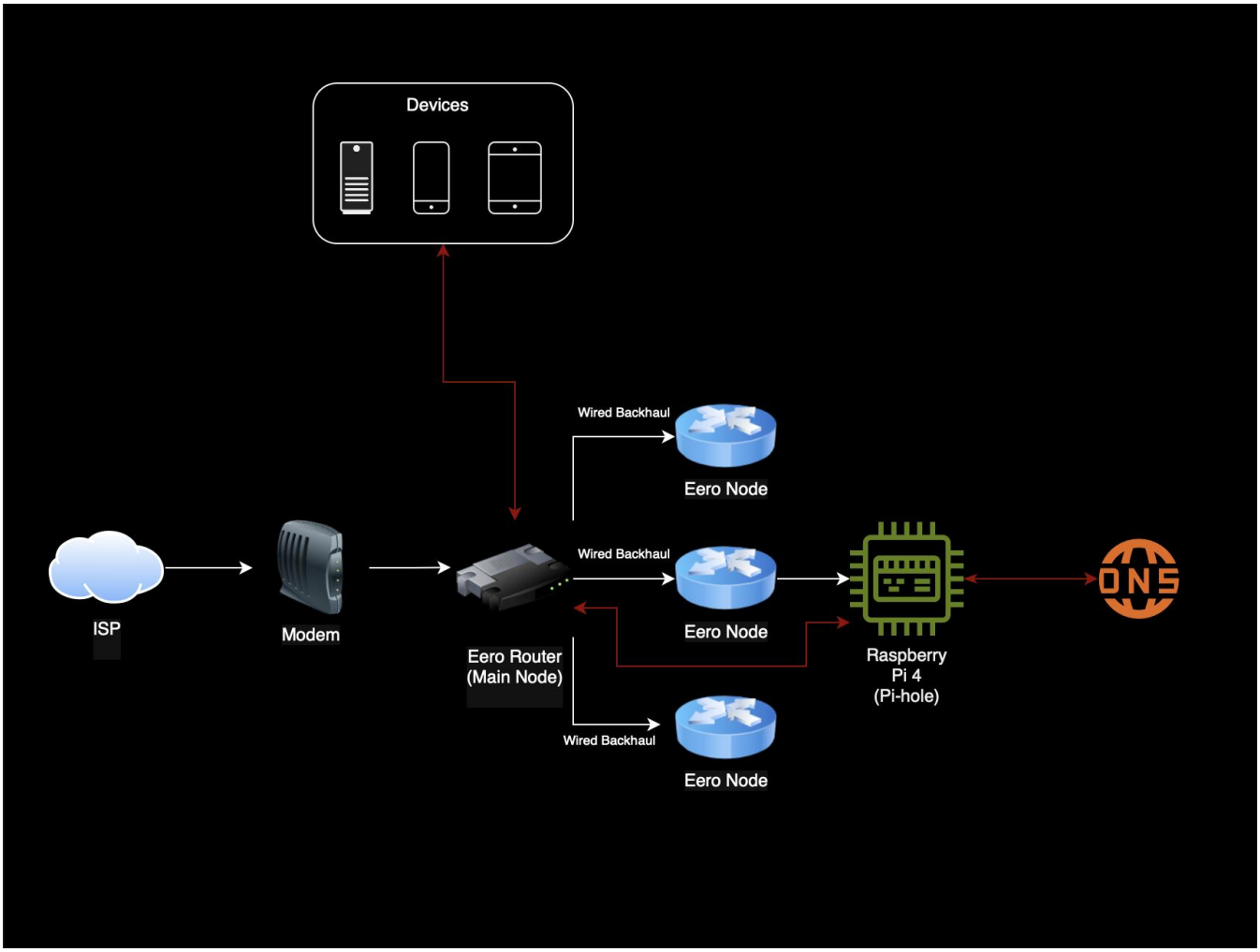
## Web Admin Dashboard Access

- Accessed via browser internally at

  `http://192.168.4.108/admin`
- Logged in using the default admin password and immediately changed it for security
- Web UI now fully operational for monitoring, blocking, and DNS configuration

# Network Topology

## Network Layout Diagram



# Network Topology

## Device Flow Explanation

The diagram above illustrates how DNS and network traffic flows within the home network. Here is a breakdown of each stage in the process:

1. **Devices (Clients):**
   Laptops, phones, tablets, and other client devices generate DNS queries when accessing websites or services.

2. **Eero Mesh Network:**
   These queries are forwarded to the main Eero router or the nearest Eero node via wired backhaul. The Eero system is configured to send all DNS requests to the Pi-hole instance instead of directly using an external DNS provider.

3. **Raspberry Pi 4 (Pi-hole):**
   Pi-hole receives and processes all incoming DNS queries. It blocks domains based on configured blocklists (e.g., adware, trackers) and forwards allowed requests upstream to the DNS provider.

4. **Upstream DNS (Cloudflare 1.1.1.1):**
   Pi-hole forwards clean queries to Cloudflare's DNS server (chosen for now). Cloudflare resolves the domain names into IP addresses and returns the responses to Pi-hole.

5. **Response Path:**
   The Pi-hole sends the final DNS response back to the originating device through the Eero network, completing the resolution process.

This setup allows Pi-hole to act as a filtering layer for all DNS activity on the network, giving visibility, control, and performance benefits without disrupting device connectivity.

# Monitoring and Maintenance

## System Health & Metrics

- **CPU Load:** 0.1% on 4 cores running 181 processes (0.7% used by FTL)
- **RAM Usage:** 3.0% of 7.6 GB is used (2.1% used by FTL)
- **Swap:** 0.0% of 512.0 MB is used
- **Kernel:** Linux 6.12.34 (Debian Bookworm, 64-bit)
- **Primary IP:** IPV4 192.168.4.108 (static)

## DNS Reply Metrics

- **Local/Cache Replies:** 35.1% (2.3M queries)
- **Forwarded Queries:** 36.2% (2.4M queries)
- **Cache Optimizer Replies:** 28.7% (1.9M queries)
- **Unanswered Queries:** <0.1% (~1.2K queries)